

Department of Computer Engineering COMSATS
University Islamabad, Attock Campus

Lab Rubrics Form	
Lab Name	Sampling and Reconstruction of Sinusoidal Signals
Lab Number	01
Submitted by:	
Student Names	Registration #
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Rubrics		Marks		
		1	2	3
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab-01

Sampling and Reconstruction of Sinusoidal Signals

Objective:

- To generate a sinusoidal signal $x(t)=\sin(2\pi f_0 t)$ with $f_0=500$ Hz.
- To sample the signal at different sampling frequencies ($f_s=800$ Hz, 1000 Hz, 2000 Hz).
- To reconstruct the sampled signals using **stem plots** in MATLAB.
- To analyze the effect of sampling frequency on signal reconstruction.
- To verify the Nyquist Sampling Theorem using spectrum plots and observe aliasing when $f_s < 2f_0$.

Lab Task:

Effect of Sampling Rate:

1: Generate a sinusoidal signal $x(t)=\sin(2\pi f_0 t)$ where $f_0 = 500$ Hz.

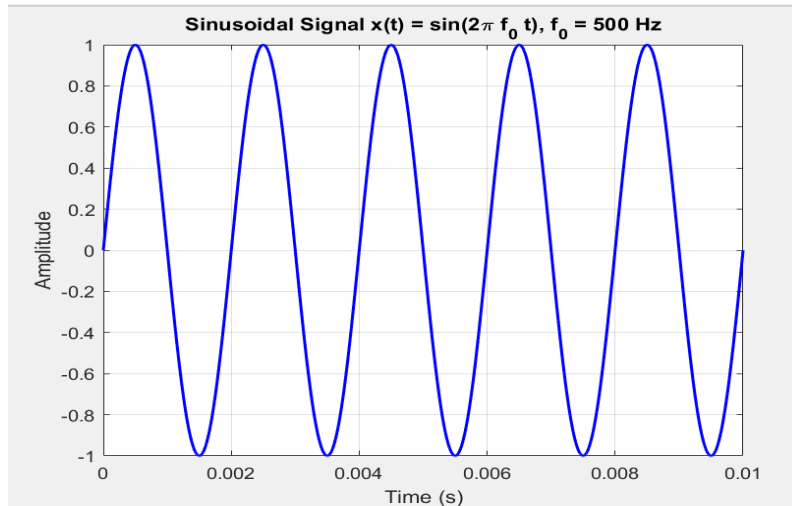
Code:

```
clc;
clear;
close all;

% Parameters
f0 = 500;           % Signal frequency (Hz)
t = 0:1/100000:0.01; % Time vector (high resolution)
x = sin(2*pi*f0*t); % Sinusoidal signal

% Plot
plot(t, x, 'b', 'LineWidth', 1.5);
xlabel('Time (s)');
ylabel('Amplitude');
title('Sinusoidal Signal x(t) = sin(2\pi f_0 t), f_0 = 500 Hz');
grid on;
```

OUTPUT:



2. Sample it at different rates: $f_s=800$ Hz, 1000 Hz, 2000 Hz.

Code:

```
clear;
close all;
f0 = 500;      % Frequency of sinusoid (Hz)
duration = 0.01; % Duration of signal (10 ms)

% Sampling frequencies
fs1 = 800; % Below Nyquist
fs2 = 1000; % At Nyquist
fs3 = 2000; % Above Nyquist

% Time vectors
t1 = 0:1/fs1:duration;
t2 = 0:1/fs2:duration;
t3 = 0:1/fs3:duration;

% Sampled signals
x1 = sin(2*pi*f0*t1);
x2 = sin(2*pi*f0*t2);
x3 = sin(2*pi*f0*t3);

% Round very small values to zero (to remove ±1e-15 noise)
x1 = round(x1, 6);
x2 = round(x2, 6);
x3 = round(x3, 6);

% Plotting sampled signals
figure('Name','Sampling of a 500 Hz Signal','NumberTitle','off');
% fs = 800 Hz
subplot(3,1,1);
stem(t1*1000, x1, 'filled', 'MarkerSize', 5);
```

```

title('Sampled Signal at f_s = 800 Hz (Aliased)');
xlabel('Time (ms)');
ylabel('Amplitude');
grid on;
xlim([0, duration*1000]);
ylim([-1 1]); % Fix y-axis range

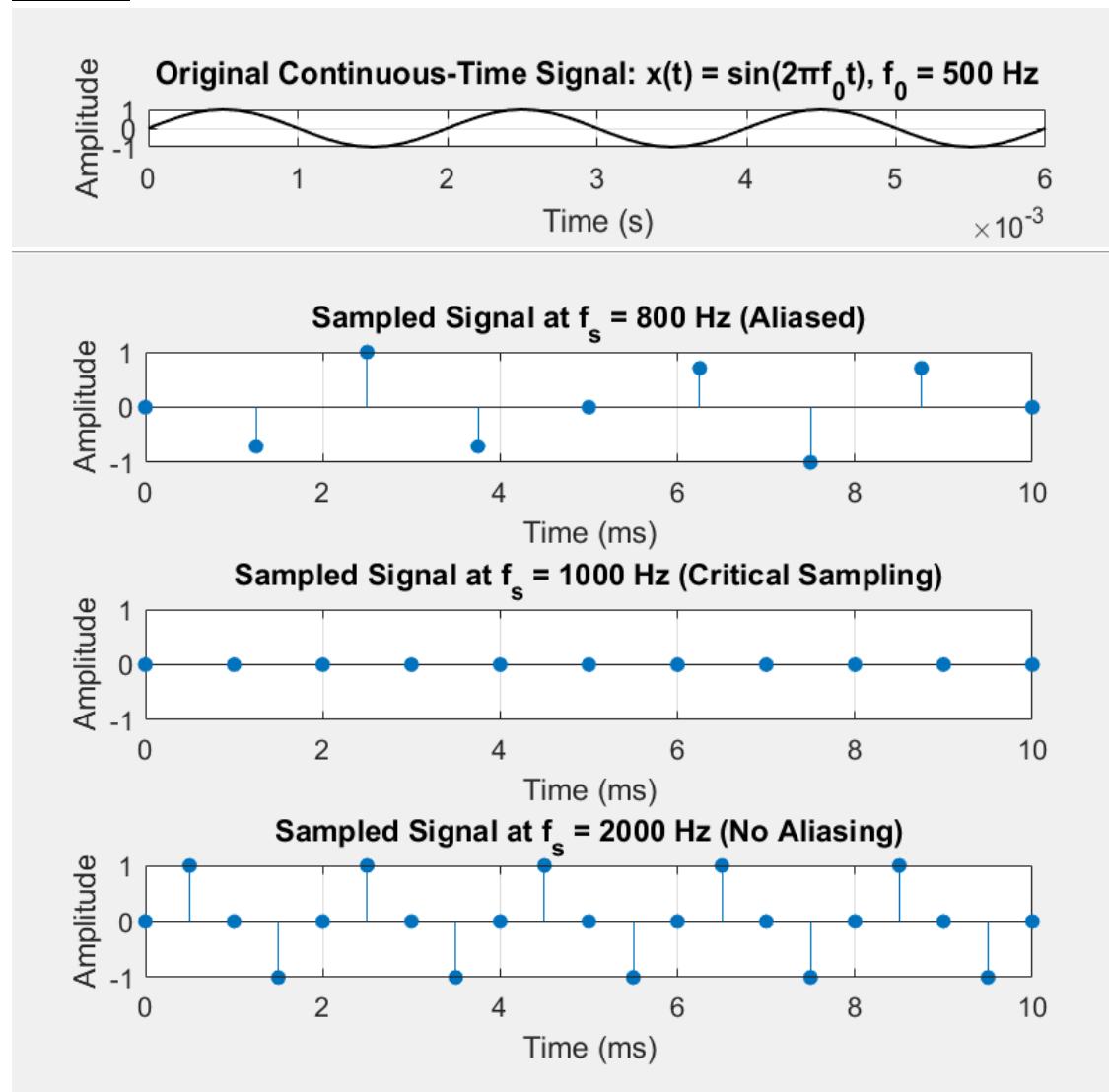
% fs = 1000 Hz
subplot(3,1,2);
stem(t2*1000, x2, 'filled', 'MarkerSize', 5);
title('Sampled Signal at f_s = 1000 Hz (Critical Sampling)');
xlabel('Time (ms)');
ylabel('Amplitude');
grid on;
xlim([0, duration*1000]);
ylim([-1 1]);

% fs = 2000 Hz
subplot(3,1,3);
stem(t3*1000, x3, 'filled', 'MarkerSize', 5);
title('Sampled Signal at f_s = 2000 Hz (No Aliasing)');
xlabel('Time (ms)');
ylabel('Amplitude');
grid on;

xlim([0, duration*1000]);
ylim([-1 1]);

```

OUTPUT:



3: Reconstruct the signal using stem and plot in MATLAB.

Code:

```
clc;
clear all;
close all;
f0 = 500; % Frequency of the sinusoid (Hz)
duration = 0.01; % Duration of the signal in seconds (10 ms)

fs1 = 800; % 800 Hz
fs2 = 1000; % 1000 Hz
fs3 = 2000; % 2000 Hz
```

```

% Create time vectors for sampling
t1 = 0:1/fs1:duration; % 800 Hz sampling
t2 = 0:1/fs2:duration; % 1000 Hz sampling
t3 = 0:1/fs3:duration; % 2000 Hz sampling

x1 = sin(2*pi*f0*t1); % Sampled signal at 800 Hz
x2 = sin(2*pi*f0*t2); % Sampled signal at 1000 Hz
x3 = sin(2*pi*f0*t3); % Sampled signal at 2000 Hz

% time vector for reconstruction
t_recon = 0:1/100000:duration;
% Reconstruct signals using sinc interpolation
x1_recon = zeros(size(t_recon));
x2_recon = zeros(size(t_recon));
x3_recon = zeros(size(t_recon));

% Reconstruction for 800 Hz sampling
for i = 1:length(t_recon)
    x1_recon(i) = sum(x1 .* sinc(fs1*(t_recon(i) - t1)));
end
% for 1000 Hz sampling
for i = 1:length(t_recon)
    x2_recon(i) = sum(x2 .* sinc(fs2*(t_recon(i) - t2)));
end
% for 2000 Hz sampling
for i = 1:length(t_recon)
    x3_recon(i) = sum(x3 .* sinc(fs3*(t_recon(i) - t3)));
end

figure;
% Original signal for comparison
subplot(4,1,1);
plot(t_recon*1000, sin(2*pi*f0*t_recon), 'b', 'LineWidth', 1.5);
title('Original Continuous-Time Signal (f_0 = 500 Hz)');
xlabel('Time (ms)');
ylabel('Amplitude');
grid on;
xlim([0, duration*1000]);

% Reconstructed from 800 Hz sampling
subplot(4,1,2);
plot(t_recon*1000, x1_recon, 'r', 'LineWidth', 1.5);
hold on;
stem(t1*1000, x1, 'k', 'filled', 'MarkerSize', 3);

```

```

title('Reconstructed Signal from f_s = 800 Hz Sampling');
xlabel('Time (ms)');
ylabel('Amplitude');
grid on;
xlim([0, duration*1000]);
legend('Reconstructed', 'Samples');

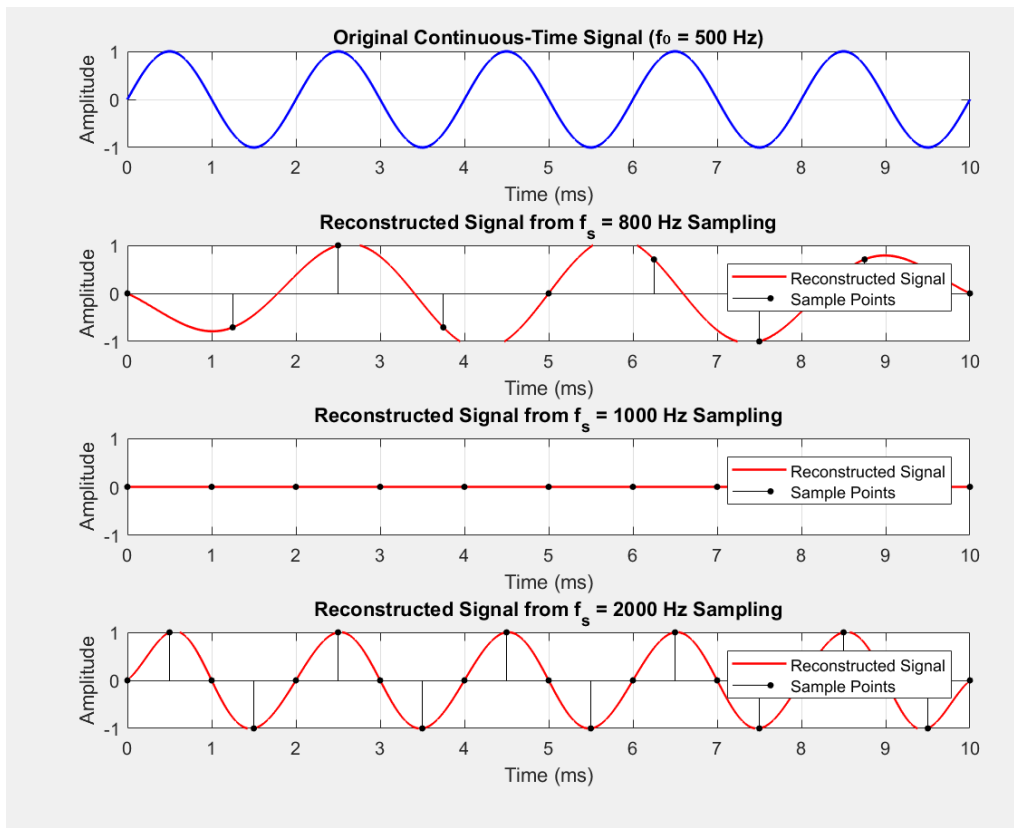
% Reconstructed from 1000 Hz sampling
subplot(4,1,3);
plot(t_recon*1000, x2_recon, 'r', 'LineWidth', 1.5);
hold on;
stem(t2*1000, x2, 'k', 'filled', 'MarkerSize', 3);
title('Reconstructed Signal from f_s = 1000 Hz Sampling');
xlabel('Time (ms)');
ylabel('Amplitude');
grid on;
xlim([0, duration*1000]);
legend('Reconstructed', 'Samples');

% Reconstructed from 2000 Hz sampling
subplot(4,1,4);
plot(t_recon*1000, x3_recon, 'r', 'LineWidth', 1.5);
hold on;
stem(t3*1000, x3, 'k', 'filled', 'MarkerSize', 3);
title('Reconstructed Signal from f_s = 2000 Hz Sampling');
xlabel('Time (ms)');
ylabel('Amplitude');
grid on;

xlim([0, duration*1000]);
legend('Reconstructed', 'Samples');
set(gcf, 'Position', [100, 100, 800, 800]);

```

OUTPUT:



Combined Result:

